

Chaitanya Kumar

Software Engineering Undergraduate

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Summary

Results-driven Computer Science undergraduate specializing in Artificial Intelligence and Full-Stack Web Development. Proven ability to build and deploy end-to-end applications—from real-time computer vision models to responsive web platforms—with a strong focus on writing clean, optimized code to solve complex engineering challenges.

Education

SRM Institute of Science and Technology (NCR Campus)

Expected: 2028

Bachelor of Technology in Computer Science and Engineering (Core)

Ghaziabad, UP

– **Current CGPA: 9.4 / 10.0**

Skills

Languages: C/C++, Python, SQL, HTML/CSS, JavaScript

Frameworks & Libraries: YOLOv8, Flask, React.js, Tailwind CSS, Scikit-Learn, Pandas, NumPy

Tools & Platforms: Git, GitHub, Netlify, Vite, OpenCV

Projects

Optimus – AI-Powered Surveillance System | *YOLOv8, OpenCV*

[GitHub Link](#)

- Engineered a real-time computer vision system using a custom-trained YOLOv8 Nano model to autonomously detect prohibited items (mobile phones) in restricted academic environments.
- Trained the deep learning model on a custom dataset using Kaggle T4 GPUs over 50 epochs, achieving a peak precision of 97.8% and a highly accurate confusion matrix.
- Developed a low-latency inference pipeline utilizing OpenCV and integrated the Telegram Bot API to instantly transmit real-time photo evidence and alerts to authorized personnel.
- Implemented an intelligent anti-spam cooldown logic to regulate alert frequency and optimized the continuous video-feed processing for efficient local execution.

IPL Match Predictor 2026 (Full-Stack ML App) | *Python, Flask*

[GitHub Link](#)

- Developed and deployed a machine learning web application predicting T20 cricket match win probabilities using a Random Forest Classifier trained on historical datasets.
- Engineered a custom 'Weighted Impact Score' algorithm (normalizing 1 Wicket to 20 Runs) to resolve scaling inconsistencies and eliminate extreme outlier predictions.
- Implemented an 'AI Insight Engine' that dynamically generates context-aware reasoning for the model's predictions based on recent team batting and bowling forms.
- Designed a fully responsive, modern Glassmorphism UI integrating dynamic requestAnimationFrame data counters for a premium user experience.

Kronix – Modern Web Application | *React.js, Vite, Tailwind CSS*

[GitHub Link](#) | [Live Link](#)

- Engineered and deployed a highly responsive, high-performance frontend web application utilizing React.js and Vite.
- Successfully configured the deployment pipeline and resolved complex build routing architectures to host the application live on Netlify.
- Designed a clean, modern user interface leveraging Tailwind CSS for an optimized user experience across various devices.

Achievements

- **Academic Excellence:** Consistently maintaining an outstanding CGPA of 9.4/10.0, reflecting strong dedication to core computer science subjects.